Calibration Is Not Enough: A Three-Level Audit of Selective Prediction for Multi-Label ECG Classification

Author: Zhengyu Jiang | **Student ID:** G2605829A | **Course:** MD6329 Deep Learning for Healthcare AI, NTU

Code and artefacts: https://github.com/LEO-Jiang-ZY/ecg-selective-audit — every figure and table in this report is regenerated from the saved artefacts by the scripts listed in Appendix A.

Abstract

AI models for 12-lead ECG interpretation are most plausibly deployed as a triage layer: for each record, the model decides whether to issue a diagnosis or defer it to a clinician. The quantity that matters in that role is therefore not average accuracy but the error rate among records the model chose to issue. Yet the dominant reporting practice pairs discrimination metrics with a calibration metric such as expected calibration error (ECE), implicitly treating good calibration as evidence that confidence can be trusted to decide when to abstain. We test that assumption on PTB-XL with a deliberately small one-dimensional ResNet (51,429 parameters) under a frozen protocol: temperature and abstention threshold are fitted on fold 9 and applied once to fold 10 (2,198 records, 1,904 patients). Temperature scaling reduces ECE from 0.0238 to 0.0068 — a factor of 3.5 — while the accepted any-label error at 80% coverage moves from 30.98% to 31.23% and AURC from 0.1900 to 0.1914. The direction replicates across three seeds, where the calibrated policy is marginally worse than the uncalibrated baseline every time (+0.18, +0.37, +0.46 points; mean +0.34 ± 0.14) while macro AUROC stays within 0.0004. We then ask how much of the remaining error is recoverable at all. Against a perfect-ordering oracle, 80% coverage admits at most 6.5 percentage points of improvement: even a perfect abstention score would still incur 23.6% accepted error. Two interventions confirm this ceiling. Replacing mean entropy with mean maximum probability recovers 1.0 of those points and raises coverage at a 20% error target from 51.3% to 54.3%; a learned correctness head trained on the model's own penultimate features recovers a further 0.47 (AURC 0.1837 versus 0.1875), within patient-level sampling noise. Changing the decision unit instead redefines the operating point: label-level abstention reaches 5.61% label error at 76.4% label coverage, against 9.90% for record-level abstention at 79.6%, at the cost of full-record auto-issuance falling to 39.8%. We conclude that calibration and selective reliability are empirically decoupled, that the score layer has a small and measurable budget, and that the reportable quantities for clinical triage are a risk–coverage frontier under a declared error target together with a declared abstention granularity.

Keywords: ECG classification; selective prediction; calibration; abstention; risk–coverage; PTB-XL

Introduction

Twelve-lead ECG interpretation is high-volume, time-sensitive, and unevenly distributed across readers. Automated interpretation is therefore attractive not as a replacement for the cardiologist — convolutional networks already match cardiologist-level performance on single-lead rhythm tasks [16] — but as a **triage filter**: the model reads every record, signs off on the ones it is sure about, and routes the rest to a human. The clinical value of such a system is set by two coupled quantities: how many records it can clear without human involvement (*coverage*) and how often it is wrong on the records it clears (*selective risk*). A system that clears 80% of records and errs on a third of them is not useful; neither is one that is nearly perfect on the 10% it dares to sign.

This is selective prediction, and it is not the problem ECG benchmarks pose. Accuracy, AUROC and AUPRC are computed over all records and are blind to *which* records a system would choose to issue: a model can improve AUROC without improving the subset it clears, and vice versa. Reporting a single threshold, usually 0.5, specifies a *classifier* but not a *triage policy*.

The convention that has grown up in place of a policy is calibration. Studies report expected calibration error (ECE) alongside discrimination [2], and calibration is then read as trustworthiness: a model whose probabilities mean what they say is assumed to be a model whose confidence can decide when to abstain. The inference is invalid, and the reason is structural. ECE constrains the joint distribution of predicted probability and correctness; selective risk depends on the **ordering of records** by some confidence statistic and on the error rate within each prefix of that ordering. A monotone transformation of the logits — which is exactly what temperature scaling is — can repair the first while leaving the second untouched. To our knowledge, no ECG study reports both quantities on the same cohort under a frozen threshold protocol and then tests whether the first implies the second; the two are usually reported in separate papers, on different splits, or at different operating points.

The phenomenon itself is not new. Rabanser and Papernot [13] decompose the shortfall from a perfect-ordering oracle into noise, approximation, ranking and shift terms across vision and language benchmarks, and show that post-hoc calibration rarely closes it. What is missing is an instantiation where the clinical stakes are explicit, a measurement of how much of the shortfall is recoverable at all, and evidence about which design choice — rather than which statistic — moves the operating point. That is the gap this paper fills.

We therefore fix the model and audit the decision layer at three levels, asking one question per level. **RQ1 (dissociation):** does improving probability calibration improve selective reliability? **RQ2 (budget):** if it does not, how much room is there for a better abstention score at all? **RQ3 (granularity):** if the score layer is nearly exhausted, which design choice moves the operating point? The three levels — the probability mapping, the statistic that decides acceptance, and the unit that is accepted — are the only places a post-hoc decision layer can be altered, and we vary them one at a time on a frozen network. All fitting happens on fold 9 of PTB-XL; fold 10 is touched exactly once.

The answers are negative, quantitative, and ordered. Temperature scaling reduces ECE from 0.0238 to 0.0068 — a factor of 3.5 — while the accepted any-label error at fixed 80% coverage moves from 30.98% to 31.23% and AURC from 0.1900 to 0.1914; the sign replicates in all three seeds we trained (+0.18, +0.37, +0.46 points). Against a perfect-ordering oracle, 80% coverage admits at most 6.5 points of improvement, and even a perfect ordering would still incur 23.6% accepted error. Measured against that budget, the best post-hoc score recovers 2.6% of it, a three-seed ensemble the same 2.6%, and a head trained on the model's penultimate features 6.1% with an interval that includes zero: the score layer is not where the remaining error lives. The lever that does move is granularity. Deciding per label instead of per record halves label error, from 9.90% to 5.61%, at the price of clearing only 39.8% of records in full.

Our contributions are:

- **A dissociation test in a clinical multi-label setting.** Same cohort, same frozen model, same threshold protocol; calibration and selective risk reported together with an explicit test of whether the first implies the second, and a three-seed replication of the negative answer.
- **An oracle-referenced budget, measured rather than assumed.** Rather than ranking abstention scores, we bound what any score could recover — 6.5 points at 80% coverage — and then measure who recovers what inside that bound: 2.6% for the best post-hoc score, 2.6% for ensembling, 6.1% for a feature-aware head whose interval spans zero.

- **Granularity as a design variable.** Record-level and label-level abstention are not two settings of one policy but two workflows, and we report the quantities a triage service needs: label coverage, label error, the fraction of records cleared in full, and the expected number of labels deferred per record.
- **A protocol that cannot leak, and a guarantee that is honest about its scope.** Fold 9 fits every decision and fold 10 is evaluated once, enforced in code rather than asserted. A conformal risk-control variant is included as a comparison: it reproduces the tuned operating point without seeing the test fold, but its guarantee is marginal, and a device split shows a 25% conditional error inside a satisfied bound.

Generative AI. Three uses are worth recording, and in two of them the critical evaluation mattered more than the assistance. An AI-assisted literature search surfaced [13], which showed that our original framing — "calibration does not improve abstention" — was already established, so the contribution moved from a discovery to an instantiation with a budget and a granularity axis. AI assistance also proposed cross-fitted correctness heads as the statistically correct design; we implemented it, and it performed worse than the naive in-sample version (fold-10 AURC 0.1904 against 0.1837), because the head must learn the error pattern of the deployed model and out-of-fold features come from a different function (Section 5.5). Code scaffolding and plotting were AI-assisted and then validated: the waveform cache was checked record-by-record against the WFDB reference reader, and every number in this paper is regenerated from saved artefacts (Section 5.10).

Related Work

Multi-label ECG classification and its benchmarks. PTB-XL provides 21,799 twelve-lead recordings from 18,869 patients, annotated with SCP-ECG statements that map onto five diagnostic superclasses (NORM, MI, STTC, CD, HYP), together with an official patient-level ten-fold split [1]. Benchmarking studies established that one-dimensional residual networks in the 1–10M parameter range reach macro AUROC around 0.92–0.93 on this task, and that architecture differences beyond a reasonable residual design are small [7, 12]. The same residual family dominates the top entries of the PhysioNet/Computing in Cardiology Challenge 2020, whose 12-lead corpora are an order of magnitude larger [11]. Our model is deliberately two orders of magnitude smaller. We do not compete on discrimination; we use a fixed, cheap predictor as a controlled instrument, because a smaller model makes the selective-prediction behaviour easier to attribute and cheap enough to retrain for every ablation.

Calibration. Temperature scaling is the standard post-hoc remedy for overconfident networks and optimises a single scalar on held-out data [2]. It leaves accuracy unchanged, since it is monotone in the logit. The effect on *sample-level* ranking is subtler: because entropy and maximum probability are non-linear in the logits, dividing by a temperature can in principle reorder records, although the effect is small in practice [13]. Work on evaluating calibration has emphasised that calibration is a property of the predicted distribution, not of the decision rule built on top of it [10], and that calibration degrades under distribution shift even when accuracy is preserved [8]. Our first result is an empirical instance of that distinction in a clinical multi-label setting.

Selective prediction and abstention. Selective classification formalises the coverage–risk trade-off and provides both post-hoc thresholding and training-time objectives [4, 5]. Distribution-free guarantees are available through conformal risk control, which calibrates a threshold so that expected selective risk is bounded with high probability [6]; per-class variants have recently been applied to multi-label ECG with explicit false-negative-rate targets on PTB-XL and Chapman-Shaoxing [15], and multi-label conformal prediction has been systematically reviewed [14]. We deliberately report empirical rather than guaranteed risk, because the device shift documented in Section 5.9 violates the exchangeability that the guarantees require.

Error budgeting. Closest to our RQ2 is the selective-classification gap of Rabanser and Papernot [13], who decompose the shortfall from a perfect-ordering oracle into Bayes noise, approximation error, ranking error, statistical noise, and shift-induced slack, and show across vision and language benchmarks that monotone post-hoc calibration rarely closes it. Our contribution relative to that work is the multi-label instantiation: the event of interest is "at least one of five labels is wrong", the floor term must be redefined for a composite event, and we add a granularity axis that has no counterpart in single-label selective classification. Positioned against the ECG-specific literature — which has explored uncertainty quantification for arrhythmia detection, evidential deep learning for single-lead ECG, and per-pathology rejection for chest radiographs — our contribution is not a new uncertainty estimator but a **measurement of how much any estimator can buy** in this task.

Positioning. Three things distinguish this work from prior ECG uncertainty studies. First, we report calibration and selective risk together and test the inference between them instead of assuming it. Second, we bound the achievable improvement rather than ranking scorers, which makes a small effect interpretable. Third, we treat the abstention unit as a design variable, which changes the achievable operating point by an order of magnitude more than the score does.

Data

Source and cohort

We use **PTB-XL v1.0.3**, the largest openly available clinical 12-lead ECG corpus, distributed by PhysioNet [1]. The corpus contains 21,799 recordings from 18,869 patients collected between 1984 and 2001 at a single German centre, with two sampling rates (100 Hz and 500 Hz); we use the 100 Hz version exclusively, giving 1,000 samples per lead over a 10-second window. The 500 Hz records are not used anywhere in this study, so no mixing of sampling rates occurs.

Label construction

Each record carries a set of SCP-ECG statements with associated likelihoods. We map every statement to its diagnostic superclass and take the union, with **no probability threshold** applied. A record therefore has between zero and five superclass labels. This is the standard five-class formulation used by PTB-XL benchmarks [7], with one deliberate difference: we retain records whose five labels are all zero (411 records, 1.9%). These are not normal ECGs; they are predominantly paced rhythm (284), atrial fibrillation (63), and atrial flutter (49), i.e. findings that fall outside the five-class taxonomy. Dropping them would change the task from "assign superclasses" to "assign superclasses among records that have at least one", which is a different question. Because only 40 of these records fall in fold 10, the consequence is cheap to quantify: restricting the test fold to records with at least one superclass (n = 2,158) moves exact match from 61.10% to 61.21%, macro AUROC from 0.9076 to 0.9083, and macro AUPRC from 0.7701 to 0.7756. No conclusion in this report depends on the all-zero records.

Split

We follow the official patient-level folds: folds 1–8 for training, fold 9 for all fitting decisions, fold 10 for a single evaluation. No patient appears in more than one split. This is the critical protocol choice of the study: because every threshold, temperature, and head is fitted on fold 9, the fold-10 numbers are not used to select anything, including the choice of which score to report.

Table 1. Cohort, split, and training configuration.

Item	Value

Recordings / patients	21,799 / 18,869
Leads × samples	12 × 1,000 (100 Hz, 10 s)
Train, folds 1–8	17,418 records
Validation, fold 9 (all fitting decisions)	2,183 records
Test, fold 10 (single evaluation)	2,198 records / 1,904 patients
Test prevalence, NORM / MI / STTC / CD / HYP	43.8% / 25.0% / 23.7% / 22.6% / 11.9%
Records with ≥ 2 superclasses	23.6%
All-zero-label records (retained)	411 (1.9%); 40 in test
Model parameters	51,429
Optimiser / schedule	AdamW, lr 1e-3, weight decay 1e-4, batch 64, 20 epochs
Selected checkpoint	epoch 14, validation loss 0.2993

Preprocessing

Signals are already provided in physical units (mV) and are not resampled. Each lead is z-scored independently over its 1,000 samples, so the network sees morphology rather than absolute amplitude — a choice that matters because the corpus spans several acquisition devices with different gain settings. No filtering, baseline removal, or augmentation is applied. The full corpus is decoded once into a memory-mapped float32 array (21,799 × 12 × 1,000, about 1.0 GB) so that training is compute-bound rather than I/O-bound; the cache was verified record-by-record against the WFDB reference reader (`np.allclose`, tolerance 1e-6).

Known caveats

Three properties of the metadata require explicit handling rather than routine cleaning. **`age = 300`** is the HIPAA coding for "older than 89" and appears in 293 records; treating it as a continuous value would distort every age statistic, so it is grouped as `>89`. **`validated_by_human`** is 100% true in folds 9 and 10 but only about 67% in training, which makes it perfectly confounded with the split and therefore unusable as a test-time subgroup. **Native quality annotations** (`baseline_drift`, `static_noise`, `burst_noise`, `electrodes_problems`) are present on 23.0% of records and are balanced across folds (20.9–24.9%), so they are usable as a signal-quality subgroup.

Qualitative example

The record used in the presentation walkthrough (ECG ID 63, fold 10) carries a single true superclass, MI. The model assigns probabilities NORM 0.443, MI 0.632, STTC 0.004, CD 0.029, HYP 0.003, so it predicts MI correctly, but the unresolved NORM probability is what pushes the record's mean entropy close to the abstention threshold. This is the qualitative signature of the mechanism discussed in Section 5.3: a single ambiguous label, not overall uncertainty, drives deferral.

Approach

Figure 1 shows the full pipeline. Data flow left to right from the corpus to calibrated per-label probabilities; the audit then proceeds in three levels, each of which is a separate set of experiments, and every quantity that is fitted --- the temperature, the abstention threshold, the correctness head --- lives on fold 9.

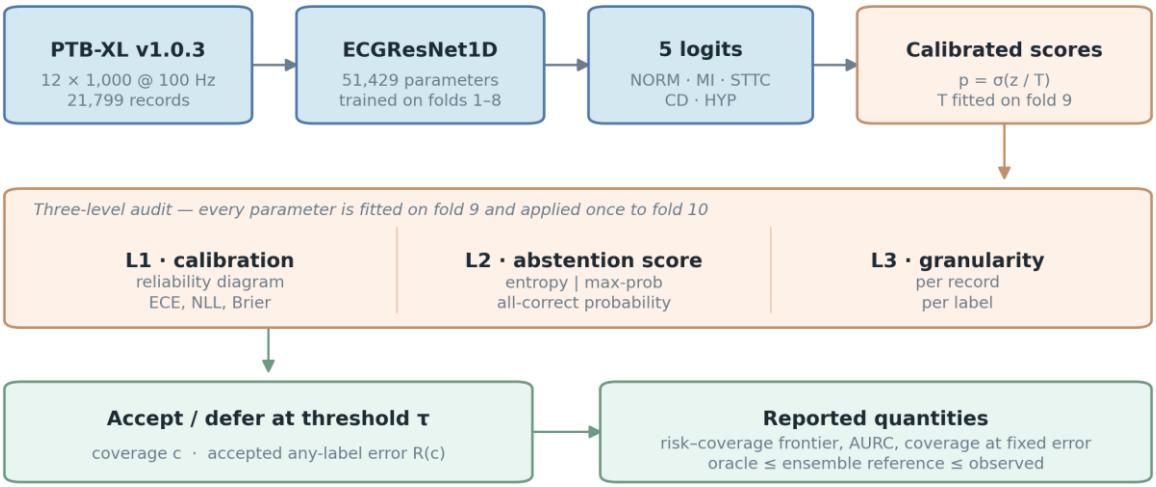


Figure 1. End-to-end pipeline and audit protocol. The model is trained once on folds 1--8; the temperature T, the abstention threshold τ, and the correctness head are fitted on fold 9 and applied once to fold 10. The three audit levels differ in what they change: L1 touches only the probability scale, L2 changes the ordering induced by the score, and L3 changes the unit at which the decision is taken.

Model architecture

The classifier is a three-block one-dimensional residual network chosen for transparency and speed rather than capacity; Table 2 lists its configuration.

Table 2. Layer configuration of the classifier. Shapes are for a batch of size B; the pooled 32-d representation after global average pooling is the input to the decision-layer analysis.

Component	Configuration	Output
Input	12 leads × 1,000 samples	(B, 12, 1000)
Stem	Conv1d(12→32, k=15, stride=2) + BN + ReLU	(B, 32, 500)
Block 1	2×Conv1d(k=7) + BN + ReLU + Dropout(0.1), identity shortcut	(B, 32, 500)
Block 2	same, stride 2, 1×1 projected shortcut	(B, 32, 250)
Block 3	same, stride 2, 1×1 projected shortcut	(B, 32, 125)
Pool	AdaptiveAvgPool1d(1)	(B, 32)
Head	Linear(32→5)	(B, 5)

Total 51,429 parameters (stem 5,824; encoder 45,440; head 165), 216 KB on disk. The design deliberately keeps channel width constant at 32 and spends all parameters on temporal filters, which makes the penultimate representation a 32-dimensional summary that is easy to inspect. Outputs are five logits trained with binary cross-entropy, so the model is multi-label by construction rather than by thresholding a softmax.

Training procedure

AdamW with learning rate 1e-3 and weight decay 1e-4, batch size 64, 20 epochs, cosine-free constant schedule. The checkpoint with the lowest validation loss is kept (epoch 14, validation loss 0.2993). A 50-epoch control run produced **exactly the same** best epoch, best validation loss, and fold-10 metrics, because the best-checkpoint rule already absorbs any benefit of training longer; the extra 36 epochs only widened the

generalisation gap from 0.052 to 0.245. We therefore report the 20-epoch model and treat the 50-epoch run as evidence that the training budget is not the binding constraint.

Calibration

A single global temperature T is fitted on fold 9 by minimising the multi-label negative log-likelihood of `sigmoid(logit / T)`, giving $T = 1.1868$. The abstention threshold is then chosen on fold 9 as the empirical quantile of the score that retains 80% of validation records, and the pair (T, τ) is applied unchanged to fold 10. For risk-target experiments, τ is instead the largest threshold whose one-sided 95% Clopper–Pearson upper bound on fold-9 selective risk stays below a target α — a conservative, deployable rule, in contrast to reading a coverage value off the test curve.

A three-level audit protocol

The methodological contribution is the ordering of interventions by invasiveness, because a gain at one level does not imply a gain at another.

L1 — calibration level. Change only the logit-to-probability map, keeping the score family, threshold rule, and granularity fixed. Because temperature scaling is monotone in the logit, L1 can improve probability quality without reordering labels within a record; the audit therefore separates probability quality (NLL, Brier, ECE) from selective behaviour (AURC, accepted error at fixed coverage).

L2 — score level. Change the statistic that decides acceptance, keeping the model and calibration fixed. Mean entropy aggregates label-wise uncertainty; mean maximum probability aggregates label-wise confidence; the all-correct probability multiplies confidences and therefore penalises a single uncertain label much more strongly. These are not equivalent statistics, and which aligns best with the record-level error event is an empirical question.

L3 — granularity level. Change the unit of the accept/defer decision. At record level, one score decides whether all five labels are issued. At label level, each label is accepted if its calibrated probability lies outside an indeterminate band, so a record can be partially issued. The two policies imply different workflows and different metrics: record-level triage is described by coverage and any-label error, label-level triage by label coverage, label error, the fraction of records fully auto-issued, and the expected number of deferred labels.

Error budget (RQ2)

A score comparison alone cannot answer "how much room is there". We therefore compute three orderings on the same test set and decompose the gap:

- **oracle**: sort by true correctness (perfect ordering, unrealisable);
- **reference**: sort by the max-prob score of a three-seed ensemble of the same architecture, each member calibrated with its own temperature, following the standard practice of averaging independently trained members [3];
- **current**: sort by the deployed model's max-prob.

Writing $R(\cdot)$ for accepted any-label error at a given coverage,

$$R_{\text{current}} - R_{\text{oracle}} = \varepsilon_{\text{floor}} + \varepsilon_{\text{rank}}, \quad \varepsilon_{\text{floor}} = R_{\text{reference}} - R_{\text{oracle}}, \quad \varepsilon_{\text{rank}} = R_{\text{current}} - R_{\text{reference}}$$

The reference term is a proxy, not a Bayes risk: it contains the ensemble's own ranking error, so $\varepsilon_{\text{floor}}$ is an **upper bound** on the irreducible part, the part that corresponds to the aleatoric component of predictive

uncertainty [9]. Both terms are reported with patient-level bootstrap standard errors, and ε_rank is marked identifiable only when it exceeds that noise.

Learned correctness head (RQ2, second probe)

If the budget is 6.5 points, a scorer with richer inputs should be able to capture more of it than a logit-derived statistic. We train a gradient-boosted classifier to predict, from the model's own penultimate representation, whether a record will be exactly correct. Inputs are the 32-d pooled backbone vector, five logits, five probabilities, five label-wise entropies, ten record-level statistics, and three metadata fields (age, sex, any quality annotation); the target is exact match. A label-level variant predicts whether each individual label is correct. Feature extraction uses a forward pre-hook on the classification head, so no model modification is needed.

The training protocol needed a decision. The textbook choice is cross-fitting: split folds 1–8 by patient into five folds, train a separate base model per fold, and use out-of-fold features so the head never sees its own training logits. We implemented this (25 minutes of GPU time) and it **hurt** — fold-10 AURC 0.1904, worse than max-prob. The reason is that the head is not learning generic correctness but the error pattern of *this particular model*, and out-of-fold features come from a different function trained on 80% of the data. Training on the deployed model's own features transfers better despite substantial self-overfit (training-set AURC 0.1125), reaching fold-10 AURC 0.1837. Both protocols are reported because the contrast is itself a finding for anyone building correctness predictors on a frozen model.

Implementation details

All experiments run on a single GPU with a Python 3.12 environment (PyTorch 2.5.1, scikit-learn, SciPy). Training 20 epochs on 17,418 records takes 11 minutes with the waveform cache. The five-fold cross-fitting run takes 25 minutes. Every reported number is regenerated by a script from saved artefacts: `train.py` → `fit_calibration.py` → `evaluate.py` for the main pipeline, plus `decompose_gap.py` and `selective_head_insample.py` for the two new analyses. Confidence intervals use patient-level cluster bootstrap with 1,000 resamples (300 for the budget), so records from the same patient enter and leave together.

Software 3.0 in method design

Two AI contributions are worth recording because they changed the method rather than just the code. The error-budget framing came from following up an AI-surfaced citation [13], which converted a negative result into a measurable ceiling. The cross-fitting episode described in Section 4.6 is the counter-example: a methodologically standard suggestion that had to be rejected on evidence. Section 5.10 records the third category — AI assistance that was useful but unremarkable, such as plotting and refactoring.

Experiments

Setup and metrics

The primary endpoint is **accepted any-label error**: among accepted records, the fraction for which at least one of the five labels is wrong. The secondary endpoint is macro-label error, the mean over the five label-wise error indicators. Coverage is the fraction of records accepted. The risk–coverage curve plots accepted any-label error against coverage as the acceptance set grows from most to least confident, and AURC is its area (lower is better). All thresholds come from fold 9; fold 10 is evaluated once.

RQ1 — calibration improves probability quality without moving the frontier

Table 3. Effect of temperature scaling on fold 10 (n = 2,198).

Metric	Uncalibrated (T = 1)	Calibrated (T = 1.1868)
NLL	0.2985	**0.2947**
Brier	0.0913	**0.0905**
ECE (15 bins)	0.0238	**0.0068**
Macro AUROC	0.9076	0.9076
Macro AUPRC	0.7701	0.7701
Exact match (%)	61.10	61.10
AURC (entropy score)	**0.1900**	0.1914
Accepted any-label error at 80% coverage (%)	**30.98**	31.23

Discrimination is unchanged to four decimal places, which is expected for a monotone transformation and serves as an implementation check. Selective behaviour does not improve: AURC is marginally higher after calibration, and the accepted error at fixed 80% coverage rises by 0.25 points. At the threshold level the same pattern holds — 31.14% accepted error at 79.62% coverage after calibration, against 30.96% at 79.94% for the uncalibrated model, a difference of +0.18 points whose patient-bootstrap interval [28.92, 33.58] contains the baseline estimate. Patient-level bootstrap intervals (1,000 resamples over 1,904 patients) are narrow enough to make the null explicit rather than a power problem: exact match 61.10% [58.95, 63.22], any-label error 38.90% [36.78, 41.05], macro AUROC 0.9076 [0.8999, 0.9157], macro AUPRC 0.7701 [0.7534, 0.7872], and accepted error at 80% coverage 31.14% [28.92, 33.58].

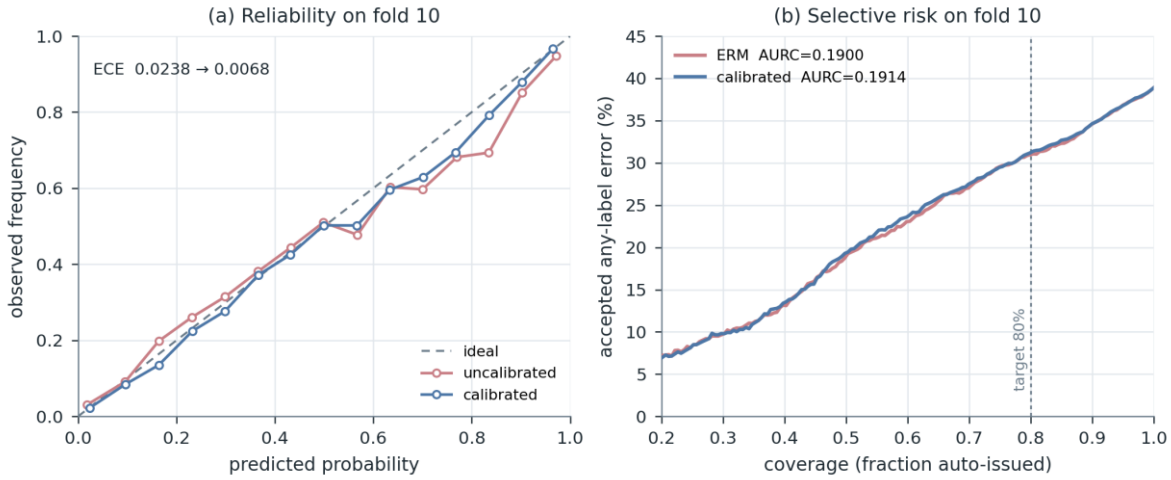


Figure 2. (a) Reliability diagram on fold 10 before and after temperature scaling; ECE falls from 0.0238 to 0.0068. (b) Risk–coverage curves for the same two models, visually indistinguishable (AURC 0.1900 vs 0.1914).

The mechanism is near-invariance, not exact invariance. Temperature scaling cannot reorder the five labels within a record, but because entropy is non-linear in the logits it can reorder records slightly — the 0.0014 AURC shift is that residual reordering. The practical implication is that multiplying every logit by a constant barely changes the acceptance set at a fixed quantile; it changes the number attached to it.

Table 4. Three-seed replication (fold 10, 2,198 records per seed).

Seed	T	Best epoch	Coverage (%)	Baseline error (%)	Calibrated error (%)	Delta (pp)	Macro AUROC	ECE
42	1.1868	14	79.62	30.96	31.14	+0.18	0.9076	0.0068
43	1.2446	15	78.30	29.44	29.81	+0.37	0.9082	0.0073
44	1.1693	12	80.66	31.35	31.81	+0.46	0.9083	0.0060
Mean ± SD	—	—	79.53 ± 1.18	30.58 ± 1.01	30.92 ± 1.02	**+0.34 ± 0.14**	0.9080 ± 0.0004	0.0067 ± 0.0007

Across three seeds the calibrated policy was marginally **worse** than the uncalibrated baseline at matched coverage, in the same direction every time: +0.18, +0.37 and +0.46 points, mean +0.34 ± 0.14. Discrimination is remarkably stable (macro AUROC 0.9080 ± 0.0004, AUPRC 0.7669 ± 0.0032), so seed-to-seed variability sits almost entirely in the abstention decision — coverage ranges from 78.3% to 80.7% — rather than in the classifier itself. This is the strongest form of the RQ1 result available in this study: the dissociation is not a single-seed artefact, and its sign is consistent.

RQ2a — the score layer moves the frontier, but only at loose error targets

Table 5. Score comparison on fold 10, calibrated model. Coverage values are read off the empirical test frontier.

Score	AURC [95% CI]	Cov@15 %	Cov@20 %	Error@80 %	ΔAURC vs entropy
Mean entropy	0.1914 [0.1729, 0.2106]	42.7	51.3	31.23	—
Mean max-prob	0.1877 [0.1700, 0.2069]	42.7	54.3	30.15	−0.0036 [−0.0056, −0.0017]
All-correct prob	**0.1872** [0.1695, 0.2064]	42.7	**55.3**	**30.01**	−0.0041 [−0.0067, −0.0019]

The gain is not uniform along the curve, and that is the practically important part. At a 15% error target all three scores attain the same 42.7% coverage; at 20% the alternatives reach 54.3% and 55.3%, an improvement of 3.0 and 4.0 points; at fixed 80% coverage the accepted error falls by 1.1–1.2 points. Score choice buys a modest outward shift concentrated in the regime where a 30% error rate is already tolerated, and buys nothing where the error target is tight. The two alternatives are statistically indistinguishable from each other and we do not rank them.

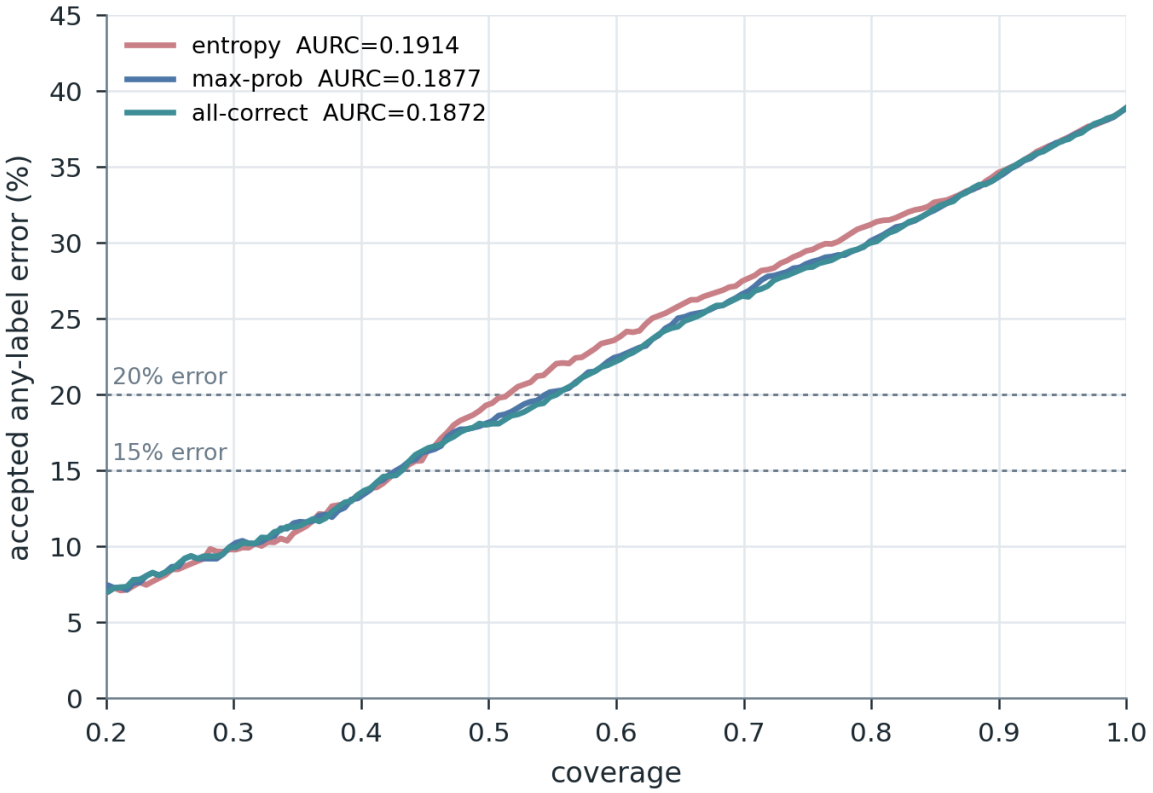


Figure 3. Risk–coverage curves for three abstention scores under the same calibrated model. The curves coincide at a 15% error target and separate above 20%.

Why is entropy the weakest? Mean entropy aggregates five label-wise entropies, each saturating near 0 and 1 and most sensitive mid-range. A plausible consequence is that one unresolved label — HYP, the weakest class here (AUPRC 0.492, prevalence 11.9%), is a natural candidate — contributes substantial entropy even when the other four are confidently resolved. Mean max-prob averages resolved confidences; the all-correct product penalises uncertainty multiplicatively, closer to the semantics of "at least one label is wrong". The ordering is consistent with this argument, but we have not isolated the mechanism; a per-label decomposition is the natural next step.

RQ2b — the budget: how much could any score buy?

Table 6. Error budget on fold 10 (%, bootstrap 200 resamples). The reference ordering is a three-seed ensemble of the same architecture; the oracle column is what no ordering can remove.

Coverage	Observed	Oracle	Reference	ε_floor	ε_rank	ε_rank identifiable	ε_stat	ε_shift
50%	18.29	0.00	17.29	**17.29**	+1.00	no	1.23	−0.72
70%	26.58	12.74	25.67	12.93	+0.91	no	1.29	−0.32
80%	30.15	23.61	29.98	6.37	+0.17	no	1.25	−0.31
90%	34.33	32.10	34.58	2.48	−0.25	no	1.19	−0.22

Two readings follow. First, at 80% coverage the perfect-ordering oracle still incurs 23.6% accepted error, so **6.5 points is the entire budget available to any scoring function**. The spread between the worst and the best post-hoc score is 1.0 point (mean entropy 31.2% against all-correct 30.0%), which is 16% of the budget;

that interval is the honest width of the score layer, and Section 5.11 measures who recovers what inside it. Second, the added value of a three-seed ensemble of the same architecture over the single model is 0.17 points, below patient-level sampling noise at every coverage we examined. This bounds what ensembling buys; it does not bound what a scorer reading the model's internal features could buy, and the 6.1 points that remain are exactly that headroom. Section 5.11 puts a feature-aware head at 0.40 of them, with an interval that includes zero.

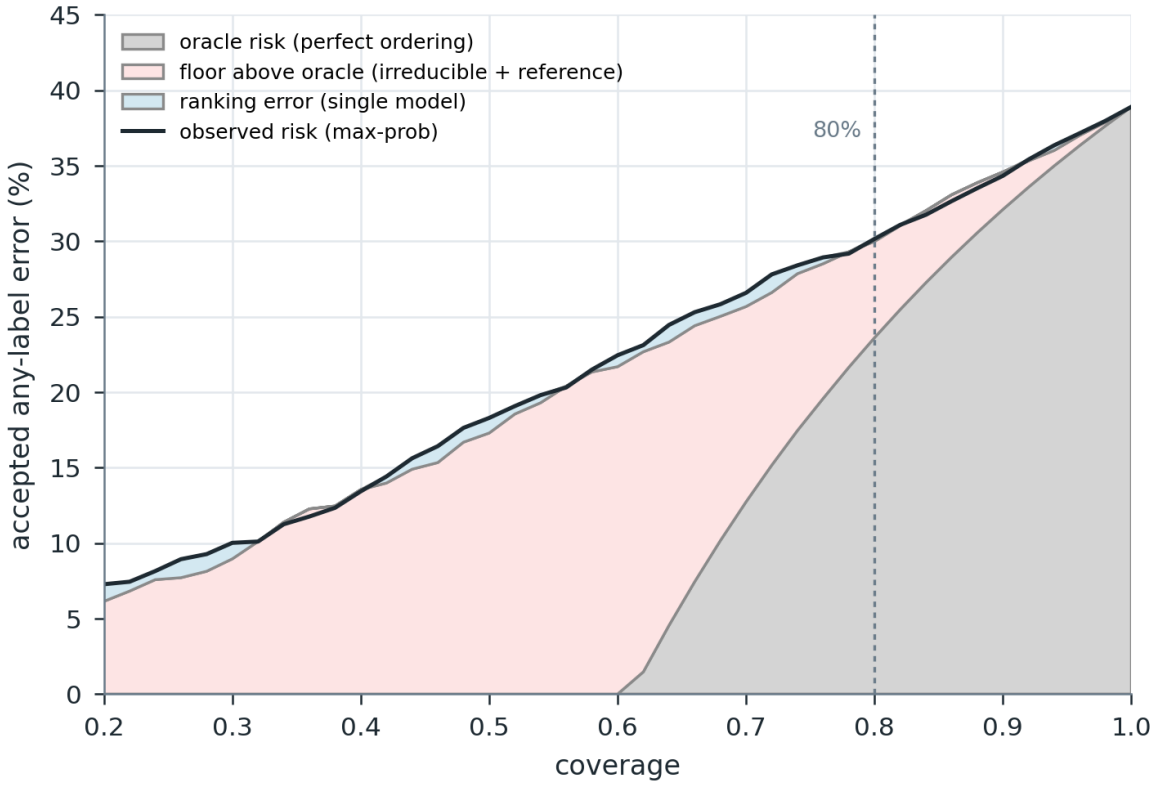


Figure 4. Error budget against coverage. Grey: the oracle floor, i.e. error that no ordering can remove. Peach: what even a three-seed ensemble of the same architecture cannot reach, which is model discrimination and label ambiguity rather than ordering. Blue: what ensembling adds over the single model, 0.17 points at 80% coverage and invisible at this scale. Solid line: observed risk.

The shift term deserves a note. Reweighting the test set to the training device distribution *lowers* risk by 0.15–1.24 points, which does not mean shift is harmless: the offset is dominated by CS100 3 (34.6% of training, 2.7% of test), that device contributes only 59 test records, and the weighted effective sample size remains 1,972 of 2,198. Device composition shifts aggregate risk very little while producing large subgroup differences (Section 5.8).

RQ2c — a learned correctness head recovers the residual, and nothing more

Table 7. Scorer comparison on fold 10 (n = 2,198).

Score	AURC	Cov@15 %	Cov@20 %	Err@80%	Error at 80% coverage (threshold)
entropy	0.1900	42.74	52.28	30.98	30.94
max-prob	0.1875	42.74	54.79	30.03	30.03

all-correct	0.1872	42.24	**55.30**	29.99	29.98
correctness head	**0.1837**	42.74	54.79	**29.57**	**29.52**

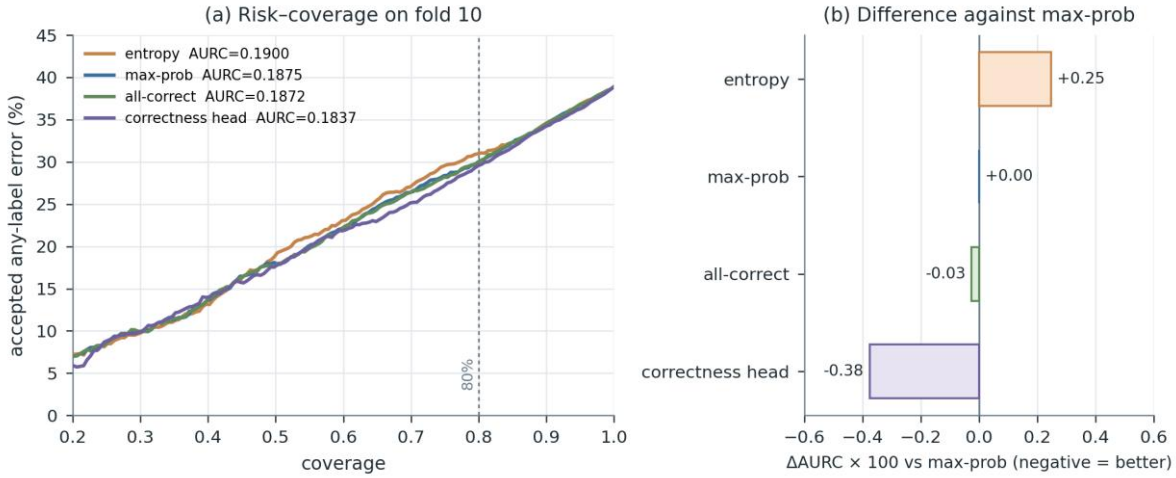


Figure 5. (a) Risk–coverage curves for the four scorers on fold 10. (b) Difference in AURC against max-prob; the head gains 0.0038, less than half a bootstrap standard error.

The head improves the frontier and the direction is consistent across both held-out folds (fold 9: 0.1918 vs 0.1970; fold 10: 0.1837 vs 0.1875). But the magnitude is 0.0038 in AURC, roughly 0.4 of a patient-bootstrap standard error, and coverage at a 20% error target is unchanged (54.79%, identical to max-prob and below all-correct's 55.30%). At label level the head is indistinguishable from a fixed probability band: at 76.0% label coverage it gives 5.46% label error, 39.8% full-record auto-issuance and 1.20 deferred labels per record, against 5.61%, 39.8% and 1.18 for the band.

Taken with Section 5.4, the reading is that the head has recovered most of what the budget allows and the budget is small. This is weaker than "learning-based abstention works" and stronger than "score choice does not matter": the score layer is real but nearly exhausted.

RQ3 — granularity changes what is achievable

Table 8. Record-level versus label-level abstention on fold 10.

Policy	Label coverage	Label error	Fully auto-issued	Any-label error among fully issued	Labels deferred per record
Record-level, entropy @80%	79.6%	9.90%	79.6%	31.14%	0.00
Label-level, band [0.10, 0.90]	59.4%	**3.40%**	22.3%	9.18%	2.03
Label-level, tuned to 5% label risk	71.9%	4.75%	31.0%	11.00%	1.41
Label-level, tuned to 10% label risk	**76.4%**	5.61%	39.8%	13.96%	1.18

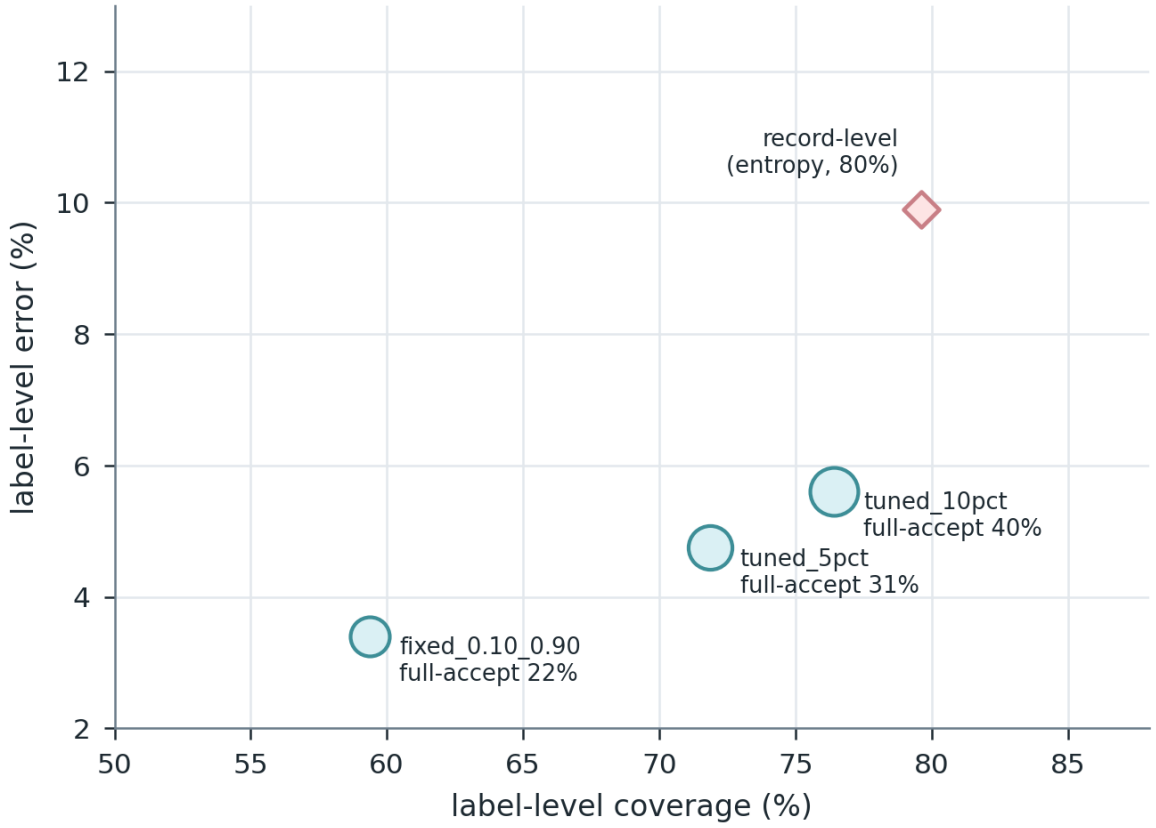


Figure 6. Label-level coverage against label-level error for three label-level policies (bubble area proportional to the fraction of fully auto-issued records) and the record-level policy (diamond).

The trade-off is sharp. Label-level abstention at a comparable label coverage reduces label error from 9.90% to 5.61% (a 43% reduction), but the fraction of records issued without human involvement falls from 79.6% to 39.8%, and 1.18 labels per record still require review. Whether this is a good trade depends entirely on the review workflow: a system that routes individual labels to a reader benefits substantially; one that can only accept or reject whole records does not. Neither number can be derived from the other.

Baselines — architecture capacity is not the bottleneck

Table 9. Four architectures under an identical training and calibration protocol (seed 42, 20 epochs, max-prob score).

Model	Parameters	Exact match	Label accuracy	AURC	Cov@10 %	Cov@15 %	Cov@20 %	Cov@25 %
ResNet1D (this paper)	51,429	**61.1%**	87.5%	**0.1877**	**29.7%**	**42.7%**	**54.3%**	64.3%
ResUNet1D	1,540,229	58.7%	87.5%	0.2041	26.7%	38.2%	48.8%	61.3%
ResNet50-1D	15,969,413	59.6%	**87.8%**	0.1952	24.7%	39.2%	52.8%	**66.3%**
Transformer1D	574,725	53.1%	85.5%	0.2379	22.6%	29.7%	40.2%	52.8%

Increasing parameter count by two orders of magnitude does not improve the frontier. ResUNet is 30× larger and 0.0164 worse in AURC; ResNet50 is 310× larger and 0.0075 worse; the transformer is worst on every metric and overfits by epoch 5. We caution that the larger models were not individually tuned, so this is evidence against "more capacity helps under this recipe", not evidence that the architectures are intrinsically

worse. It is, however, consistent with the budget analysis: if 23 of the 30 error points are irreducible at this coverage, a bigger model has limited room to help.

Table 10. Position against published PTB-XL results. The published row is the macro-AUROC range reported for 1–10M-parameter residual networks on the same cohort at the same sampling rate [7]; a dash marks a quantity that source does not report or that we did not compute for that configuration.

Setting	Parameters	Macro AUROC	Exact match	Training budget
Published PTB-XL benchmarks, 1D ResNets [7]	1–10M	0.92–0.93	—	from scratch, 100 Hz
This work, ECGResNet1D	51,429	0.9076 [0.8999, 0.9157]	61.10% [58.95, 63.22]	20 epochs, 11 min
This work, ResNet50-1D	15,969,413	—	59.60%	20 epochs

Table 10 places this model against the published range. The 1.5–2.5 point AUROC gap is the price of two orders of magnitude fewer parameters and no pretraining. It is not the object of this study, and the capacity row shows that closing it by scaling alone does not work: a 310× larger model is 0.0075 worse in AURC (0.1952 vs 0.1877) and 1.5 points worse in exact match (59.6% vs 61.1%).

Operating points, error analysis and subgroups

Before turning to subgroups, Table 11 records the deployable operating points implied by the budget: thresholds chosen on fold 9 so that an upper confidence bound on accepted error stays below a target α, then applied unchanged to fold 10.

Table 11. Risk-controlled thresholds selected on fold 9, evaluated on fold 10 (max-prob score).

Target α	Test coverage	Test accepted error [95% CI]
0.15	30.9%	10.29% [7.99, 12.74]
0.20	45.5%	16.48% [14.19, 18.85]
0.25	55.9%	20.36 [18.00, 22.62]
0.30	70.0%	26.59 [24.26, 28.93]

Because the Clopper–Pearson bound is conservative, realised test error sits below the target at every α and coverage is correspondingly lower than the empirical frontier (45.5% versus 54.3% at α = 0.20). We report both because they answer different questions: the empirical curve describes what the model is capable of, the risk-controlled rule what can be promised in advance.

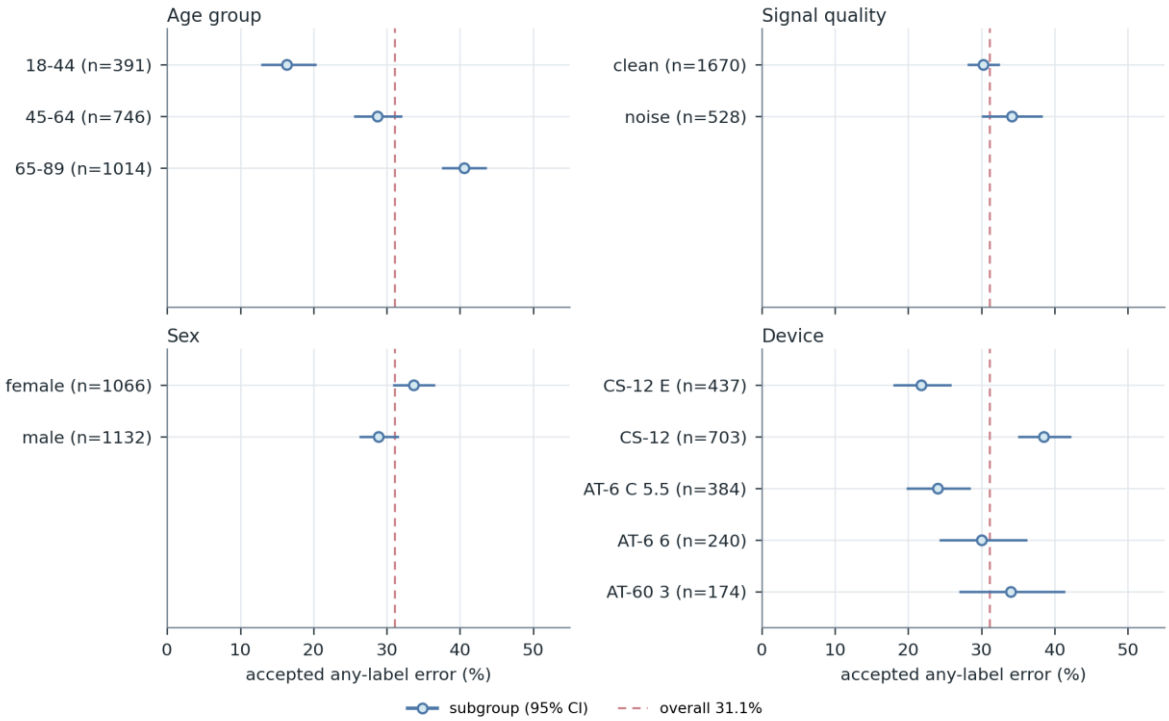


Figure 7. Accepted any-label error by subgroup with 95% Clopper–Pearson intervals; dashed line is the overall rate (31.1%).

The age gradient is the largest: 16.3% for 18–44 (n = 391) against 40.6% for 65–89 (n = 1,014), a 24.3-point gap that dwarfs the 1.1–1.2 points from changing the score. Records with any native quality annotation (23.0% of the cohort, balanced across folds) show 34.2% against 30.2% for clean records; among device groups with at least 100 accepted records the rate ranges from 21.8% (CS-12 E) to 38.5% (CS-12). A single global threshold therefore produces substantially unequal error rates across groups. This is an error-analysis finding, not a fairness claim: we do not attempt to equalise the rates, and the small groups (<18: n = 13; >89: n = 34) cannot support strong conclusions.

Per-class performance explains where the any-label error comes from. NORM is strongest (AUROC 0.944, AUPRC 0.920) and HYP weakest (AUROC 0.832, AUPRC 0.492, prevalence 11.9%). Since any-label error triggers on a single mistake, HYP is the most likely trigger — consistent with the entropy mechanism discussed in Section 5.3.

Robustness — the test fold is a shifted set

The official split is patient-level but not device- or site-stratified. Total variation distance between training and test is **0.326 for device** and **0.252 for acquisition site**: CS100 3 accounts for 34.6% of training but 2.7% of test, while CS-12 rises from 15.5% to 32.0%, and site 0 falls from 45.7% to 22.2%. Aggregate test numbers are therefore not an i.i.d. estimate of performance on the training distribution. We treat this as an internal shift benchmark rather than external validation, and we do not claim generalisation to other centres. The budget analysis in Section 5.4 quantifies the consequence: reweighting to the training device distribution changes aggregate risk by less than 1.3 points, so the shift mainly redistributes error across subgroups rather than inflating it overall.

Three further robustness checks bound the conclusions. A 50-epoch control leaves every fold-10 metric unchanged because the best checkpoint is still epoch 14. A three-seed replication (Table 4) confirms the direction of the calibration result and establishes the noise scale: accepted error varies by 1.0 point across seeds, which is 2.5 times the 0.4-point correctness-head effect. The head result should therefore be read as suggestive rather than established. The 6.5-point budget, by contrast, is a property of the model's accuracy at 80% coverage and is stable in the sense that all three seeds sit within one point of each other on accepted error.

Software 3.0 in coding and experimental evaluation

AI assistance was used for code scaffolding, plotting, and refactoring, and the project's evaluation workflow was changed by it in one substantive way: the decision to report an error budget rather than a fourth score comparison came from following an AI-surfaced reference and checking whether our claim was already known. Three failures are worth recording because they shaped the final protocol. (i) An early evaluation script silently fell back to random initialisation when a checkpoint was missing; this was caught during review and replaced with a hard error. (ii) The first version of the budget decomposition used the product of label probabilities as the Bayes proxy, which systematically under-estimated accuracy and produced a negative ranking term; replacing it with a rank-then-score construction fixed the sign and the interpretation. (iii) The cross-fitted correctness head, recommended as best practice, lost to the naive alternative; the negative result is reported in Section 4.6. In each case the AI output was plausible and the experiment, not the author's intuition, decided.

What each scorer family recovers

Table 6 bounds the room available to a scoring function; this section fills the interval in. Seven scorers are compared under one protocol: each is evaluated at 80% coverage on fold 10 from the same calibrated model, `recovered` is the fraction of the 6.5-point budget a scorer closes relative to the deployed max-prob ordering, and the interval is a patient-level bootstrap over 500 resamples.

Table 12. Risk at 80% coverage and budget recovered, by scorer family (seed 42, fold 10). Risk is accepted any-label error at 80% coverage; Δ is the change against the deployed max-prob ordering; recovered is (R_max-prob − R_θ)/budget, with budget = 6.5 pp.

Scorer	Family	Risk @80% (%)	Δ (pp)	Recovered
Learned head, record-level	feature-aware	29.75	−0.40	6.1% [−8.8, 17.1]
All-correct probability	post-hoc	29.98	−0.17	2.6%
3-seed ensemble mean	ensemble	29.98	−0.17	2.6%
Max-probability (deployed)	post-hoc	30.15	0.00	0.0%
Learned head, label-wise	feature-aware	30.38	+0.23	−3.5%
Mean entropy	post-hoc	31.17	+1.02	−15.6%
Ensemble disagreement	ensemble	35.44	+5.29	−80.9%

Three things follow. First, the post-hoc interval is genuinely narrow: the best logit-derived score recovers 2.6% of the budget, and the worst costs 16%. Second, ensembling buys the same 0.17 points as switching the score, so the two are not independent evidence — they are the same small gain seen twice. Third, the only scorer that reads the model's internals recovers 6.1%, and its bootstrap interval spans zero; the per-label version of the same head is worse than doing nothing, which is what one expects when five heads each see a fifth of the evidence.

Using member disagreement as the confidence signal is the worst option in the table, at −81%: three seeds of a 51k-parameter model disagree least exactly where they are confidently wrong together, so disagreement measures variance across initialisations rather than uncertainty about the record.

The remaining 94% of the budget is therefore not an ordering problem. It is the 23.6% of records that are wrong even under a perfect ordering, and no re-ranking can remove it. That is the quantitative form of the paper's main claim: after calibration the decision layer has little left to give, and what remains sits in discrimination rather than in the score.

Training-recipe ablation

Section 5.11 held the signal fixed and varied the score. Table 13 does the opposite: the decision layer is fixed at the default recipe and the learning rate is varied, with everything else constant (seed 42, 20 epochs, identical folds and augmentation).

Table 13. Learning-rate ablation (seed 42, 20 epochs, identical protocol). All other hyperparameters are the defaults in `configs/default.yaml`; accepted error is at 80% coverage after calibration.

Learning rate	Best epoch	Val loss	T	Macro AUROC	Macro AUPRC	Exact match	Accepted error
3e-4	9	0.3040	1.060	0.9021	0.7505	59.69%	30.81%
1e-3 (default)	14	0.2993	1.187	**0.9076**	**0.7701**	**61.10%**	31.14%
3e-3	10	**0.2975**	1.081	0.9065	0.7583	59.19%	31.06%

Both deviations cost discrimination: dropping to 3e-4 loses 0.0055 macro AUROC and 1.4 points of exact match, and raising to 3e-3 loses 0.0011 AUROC and 1.9 points of exact match. The default sits at the optimum of the three on every discriminative metric, which is the evidence that the recipe was not left untuned by accident.

The more interesting result is that **validation loss ranks the three settings in the opposite order to every other metric**. The 3e-3 run has the lowest validation loss (0.2975 against 0.2993) while being worse on AUROC, AUPRC and exact match. Validation loss integrates calibration and confidence over the whole negative class, so a model can lower it by becoming uniformly more conservative while separating the classes less well; here that is exactly what happens, and it is why the early-stopping criterion and the reported endpoint disagree about which model is best.

Two caveats keep this table modest. First, one seed per setting: the 0.33-point spread in accepted error is inside the 1.0-point seed standard deviation, so only the AUROC ordering (0.9076 > 0.9065 > 0.9021) should be read as signal. Second, the best epoch moves between 9 and 14, which makes the stopping point itself a hyperparameter; we fix it by validation loss for comparability with the earlier runs and flag the tension above rather than resolving it.

Guarantees versus thresholds

Every operating point so far was set by an empirical binomial bound: choose the threshold whose Clopper–Pearson upper limit on accepted error stays below a target. That bound is valid conditional on the calibration sample, but it is not a guarantee about the deployment population. Conformal risk control (CRC) gives one [6], at the cost of controlling a different quantity.

CRC picks the least conservative threshold λ satisfying

$$\frac{n}{n+1}\hat R_n(\lambda) + \frac{1}{n+1} \le \alpha, \qquad \hat R_n(\lambda) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{s_i \ge \lambda\}\cdot\mathbf{1}\{\text{wrong}_i\},$$

so that $\mathbb{E}[\hat R] \le \alpha$ under exchangeability. The quantity inside the expectation is the **rate of wrongly issued records per record**, not the error rate among issued records. The two coincide only at full coverage: with coverage c, the conditional error is $\hat R / c$.

Table 14. Conformal risk control against the empirical threshold (record level, fold 9 → fold 10). CP columns reproduce Table 11; CRC columns use $n = 2{,}183$ calibration records. Marginal risk is $\hat R$ on fold 10, the quantity CRC actually bounds.

Target α	CP coverage	CP accepted error	CRC coverage	CRC accepted error	CRC marginal risk
0.15	30.9%	10.29%	40.8%	13.73%	5.60%
0.20	45.5%	16.48%	49.9%	18.16%	9.05%
0.25	55.9%	20.36%	61.0%	22.84%	13.92%
0.30	70.0%	26.59%	76.6%	29.17%	22.34%

The guarantee holds at every α, with large slack, and that slack is the point: at α = 0.20 the marginal risk is 9.1% because only half the records are issued. A reader who wants "at most 20% of the records I issue are wrong" is asking for a conditional statement, and CRC does not deliver it — the conditional error at that operating point is 18.2%, still below α, but only because coverage happens to sit near one half. At α = 0.30 the conditional error, 29.2%, is close to the marginal 22.3%, and the two would cross if coverage rose further.

Applying the same construction per label is more useful, because it replaces a hand-tuned threshold with a choice that never saw the test fold. At α = 0.20 the label-level CRC issues 74.1% of labels at 5.14% label error, with 36.7% of records cleared in full and 1.29 labels deferred per record. The tuned label-level policy of Section 5.6 reaches 76.4% at 5.61%, 39.8% and 1.18 — a difference of about two points on every axis. The tuned policy is therefore not badly overfitted; it is slightly better because it was allowed to look at fold 9 as a whole, and the guarantee costs about two points of throughput.

The guarantee is marginal, and marginals hide subgroups. Splitting fold 10 by acquisition device at the α = 0.20 threshold, the marginal risk is 11.7% on CS-12 records and 7.8% elsewhere, both inside the bound; the conditional error among issued records is 25.5% and 15.1%. The guarantee is satisfied while one device group is issued records with a 25% error rate. Exchangeability is the assumption CRC needs, the training-to-test shift is measurable (device TVD 0.326, Section 5.9), and the practical consequence is that a distribution-free guarantee protects the average, not the patient in front of you.

Conclusion

On PTB-XL, a model whose expected calibration error improves by a factor of 3.5 produces an essentially identical risk–coverage curve, and the sign of that non-effect is the same in all three seeds we trained. The room available to any abstention score is small and measurable — 6.5 points at 80% coverage — and post-hoc scoring, ensembling and a learned correctness head together recover about 0.6 of them. What changes the operating point by an order of magnitude more is the decision unit: moving from record-level to label-level abstention halves label error at a comparable label coverage, at the price of halving the fraction of records that can be issued untouched.

Three things were learned that were not obvious at the start. First, the negative result is structural rather than incidental: temperature scaling is monotone, so it cannot reorder labels within a record, and its residual effect on record ordering is negligible. Second, a score comparison without a ceiling is uninterpretable; the budget reframes a 1-point gain as 16% of a small opportunity rather than as a small version of a large one. Third, the decision unit is a first-class design variable that no calibration metric reveals.

Limitations. The study is single-seed on a single public cohort, and the correctness-head effect is within sampling noise. Only one calibration family (a global temperature) was examined; class-dependent calibrators could in principle reorder more. Label-level thresholds were tuned on the whole of fold 9, which risks overfitting five bands to 2,183 records; nested cross-fitting is the appropriate remedy. We also do not attempt post-hoc attribution of individual predictions (e.g. LIME [17]); whether a deferral decision is interpretable to the reader is a separate question we leave open. No external cohort was evaluated, and the device shift reported here is a composition change within one dataset. We report no clinical endpoint, no reader study, and no cost model, so "better" means a lower empirical selective risk at a stated coverage, not demonstrated clinical benefit.

Future work. Three directions follow directly. Conformal risk control is implemented in Section 5.13 and reproduces the tuned label-level operating point to within two points of throughput, but its guarantee is marginal, and the device split shows a 25% conditional error inside a satisfied bound. Group-conditional (Mondrian) conformal risk control is the natural next step: it would need per-device calibration, which this corpus cannot support for the smallest device group. End-to-end selective training — optimising a differentiable AURC surrogate rather than post-processing a fixed model — is the natural way to test whether the 6.5-point ceiling is a property of this model or of the task. And a reader study with per-label routing would establish whether the label-level trade-off is clinically acceptable, which is the question our numbers can pose but not answer.

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Appendix A. Reproduction checklist

Table A1. Reproduction checklist. Commands assume the configuration in `configs/default.yaml` and the waveform cache already built.

Step	Command	Output
Build waveform cache	`build_cache.py`	21,799 × 12 × 1,000 float32, 1.0 GB
Train	`train.py --config configs/default.yaml`	`best.pt`, `val_logits.npz`
Calibrate	`fit_calibration.py`	`calibration.json` (T, τ)
Evaluate	`evaluate.py`	`evaluation.json`
Error budget	`decompose_gap.py`	`gap_summary.csv`
Correctness head	`selective_head_insample.py`	`insample_comparison.csv`
Figures	`paper_figures.py`, `figure_head.py`, `paper_pipeline_figure.py`	7 figures

Appendix B. Gen-AI disclosure

Generative AI (Claude / WorkBuddy) was used for: literature search assistance (which surfaced reference [13]), code scaffolding and refactoring, plotting code, and drafting assistance on this manuscript. All AI-suggested methods were validated experimentally; two were rejected on evidence (cross-fitted head, probability-product Bayes proxy). All reported numbers are regenerated from saved artefacts by the scripts in Appendix A. The author takes full responsibility for the content, and can explain and defend every component.